Q.27 What are the differences between linear and non-linear system?

Q.28 Define automatically controlled closed loop system. How it is different from manual control system.

Q.29 Explain the working of synchro-pair error detector.

Q.30 Define pole and zero of the system. Give any example of it.

Q.31 Define Routh-hurwitz criteria? Where it is uses for?

Q.32 Draw labelled diagram of stepper motor and explain its working principle along with applications?

Q.33 Define Root locus technique and listed 4 steps to draw root locus technique?

Q.34 Write the differences between signal flow graph and block diagram reduction method to find transfer function.

Q.35 Draw the block diagram of closed loop control system. Explain it basic elements.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What is servomotor? What is the working principle of it? Write down any 5 applications of it. Draw the torque-speed characteristics of servo motor.

Q.37 Define closed loop control system. Draw the block diagram of closed loop control system. Write down the 2 application of closed loop system. Listed two advantages and disadvantages of closed loop control system also.

Q.38 What is first order system? Find the time response of first order system when subjected to step input. Write any one example of it.

No. of Printed Pages : 4 181531/121531/031531
Roll No. .... /073641

3rd Sem / IC, EI

Subject:- Basics of Control System / Const. sys.

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which one of the following is not correct for the closed loop control system?
a) It doesn't have any feedback
b) It is complex in design
c) It is highly sensitive towards the change in feedback
d) Design of such circuit is usually expensive

Q.2 Mason's gain rule is used to find the_____ of the system?
a) Time response of the system
b) Frequency response of the system
c) Gain margin and phase margin of the system
d) Transfer function of the system

Q.3 In block diagram reduction technique, if two blocks are in parallel then the overall gain of the system is
a) Product of individual gain of both the blocks
b) Division of individual gain of both the blocks
c) Sum of individual gain of both the blocks.
d) Exponential of product of individual gain of both the blocks

Q.4 When all the elements of any row of Routh array becomes zero then the system becomes
a) Stable
b) Marginally stable
c) Unstable
d) May be either unstable or marginally stable

Q.5 Which of the following statement is correct about the unit step signal?

a) The magnitude is 1 for time less than 0 and zero otherwise

b) The magnitude is 1 for time greater than 0 and zero otherwise

c) The magnitude is 1 for time equal to 0 and zero otherwise

d) The magnitude does not defined at time equal to 0.

Q.6 The Bode plot is the graph between

a) The decibel magnitude and lagorithmic frequency

b) The magnitude and frequency

c) The logarithmic magnitude and logarithmic frequency.

d) The magnitude and time

Q.7 The system is said to be non-linear if

a) It follows the principle of superposition and homogeneity

b) It is time invariant

c) It does not follow the principle of superposition and homogeneity

d) None of the above

Q.8 The output voltage of tachometer in proportional to

a) The magnetic flux b) Both c and d

c) The speed d) The induced emf.

Q.9 A potentiometer is

a) An active device

b) A passive device

c) An inductive transducer

d) Used to control the flow of liquid

Q.10 The time taken by the system response to reach from 0% to 100% is known as

a) Rise time b) Peak time

c) Settling time d) Delay time

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Open loop system is more accurate than closed loop system. (True/False)

Q.12 Laplace transforms of the unit step function is_____.

Q.13 Define the peak time.

Q.14 Define DC tachometer.

Q.15 $s^3 + 3s^2 + 2s + 1 = 0$ is an unstable system. (True/False)

Q.16 Time taken by the system response to reach to 50% of the final value is called as _______.

Q.17 If the denominator of a system transfer function becomes 0 then the value of s is known as ________ of the system. (pole /zero)

Q.18 Define open loop control system.

Q.19 Define time constant of first order system when subjected to step input.

Q.20 What is potentiometer?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write any 5 differences between open loop and closed loop system.

Q.22 What are the basic elements of servo mechanism? Draw its block diagram.

Q.23 Define transfer function. How Mason's gain rule can be used to find the transfer function of the system?

Q.24 What is difference between over damped, critically damped and underdamped system? Give one example of each.

Q.25 What are the standard test signals in control system. Define mathematically.

Q.26 Draw the Bode plot of given below transfer function-

$$T.F = \frac{5(S+1)}{S(S+5)}$$